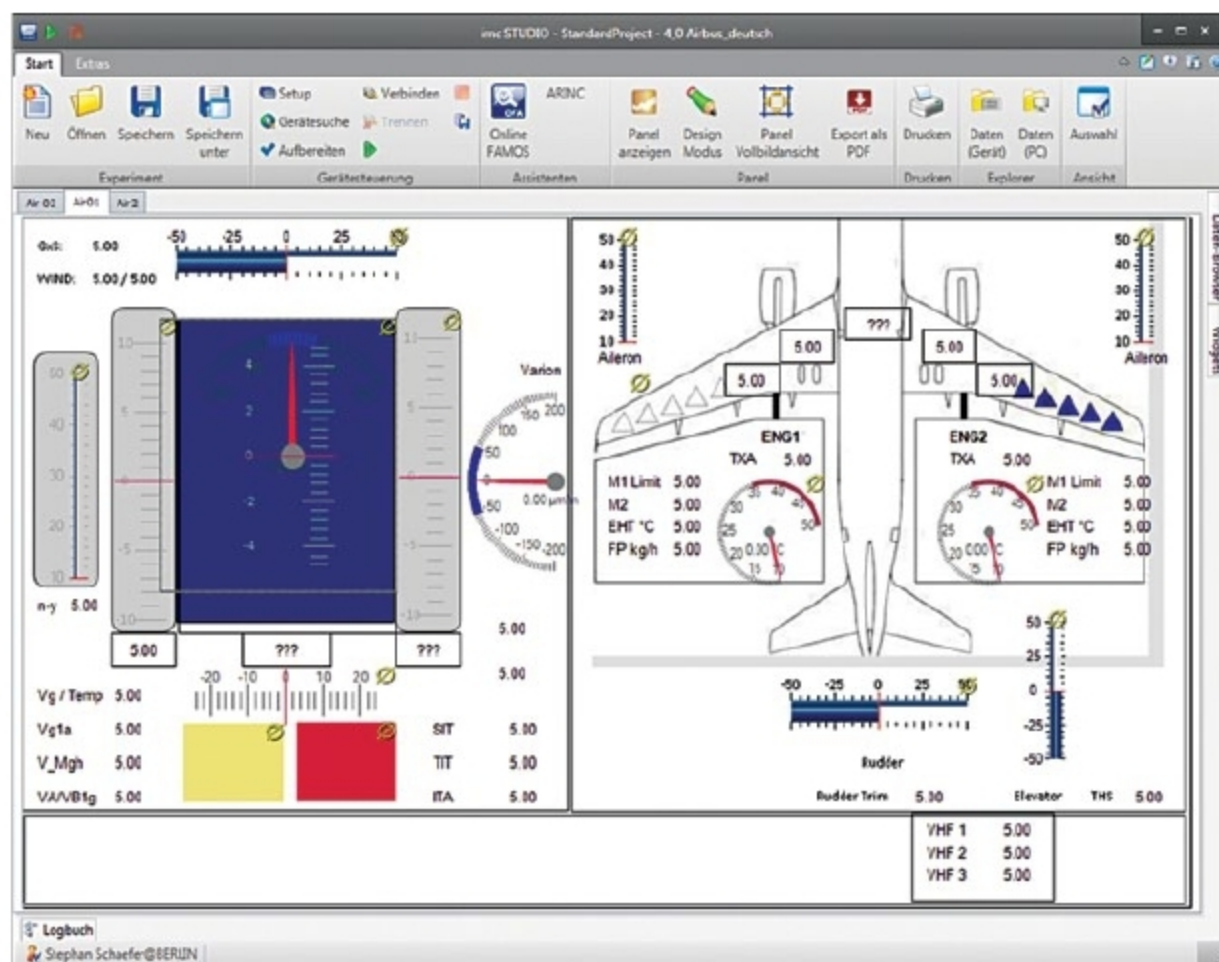




imc Studio (Credit: www.imc-tm.com)

Q.station X is a high-performance edge controller for data acquisition, monitoring and control, remote configuration and universal connectivity. It provides accurate synchronisation of measurement data, high-speed redundant data logging, and parallel communication over TCP/IP, CAN, PROFIBUS, Modbus, and EtherCAT. It comes with an optional full-featured programmable application controller designed for complex control and automation tasks.

A wide variety of signals and sensors, as well as high channel counts from digital sources and aircraft buses, are acquired over multiple channels, monitored online and continuously stored in a common data server. A flight test system based on imc's measurements and video technology (up to eight cameras) is a useful solution. All acquired analogue and digital values from a variety of sensors are stored in the aircraft and transmitted in parallel to the control station by pulse code modulation. Thus, it is possible to view and evaluate measurement data via imc Studio monitor at any time.

imc Studio offers numerous aircraft-specific indicators, such as a moving map with respect to position. These display elements can be

freely placed on panels, thus allowing custom interface designs. Panels can then be distributed on multiple monitors (up to eight). This allows the test team to have everything in view, even during complex flight testing.

Keysight's FieldFox microwave analysers are equipped to handle every task from routine maintenance

to in-depth troubleshooting. FieldFox delivers microwave measurements like spectrum analysis, vector network analysis, antenna test and so on. These analysers have a robust structure with a water-resistant chassis, keypad, and a case that withstands shock and vibrations with a temperature range from -10°C to +55°C. FieldFox meets MIL-PRF-28800F class 2 standards in harsh environments.

Naval

There are different types of measurements in ship systems, to achieve a complete overview of any condition onboard ships. It is vital to measure navigation, operation of power plant, operation of cargo system, and ensure safety of crew and equipment, and to keep a check on the weather and environment in general for everyone's safety.

For navigation, measurement of position of the ship and foreign ships' coordinates to control its own movement along the desired trajectory, position stabilisation and use of anti-collision systems is required. For measuring safe and economical operation of the ship's power plant, mechanical motion parameters of devices driving generators and engines should be measured separately.

Measurement of torque, power and revolution speed on the shaft of the main engine determines the economical operation of the ship's propulsion. Measurement of quantities like voltage, current, power, frequency, power factor, insulation resistance and total harmonic distortion characterise the power grid.

An example of such a system is SeaPerformer developed by Enamor Ltd, Gdynia, Poland, which cooperates closely with Gdynia Maritime University. The system has been successfully implemented in dozens of deep-sea ships.

To examine the acoustic fingerprint of surface and underwater vessels, acoustic signals are measured, recorded and processed to calculate spectra.



FieldFox microwave analyser (Credit: www.keysight.com)